KanvaCol Requirements

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Overview

KanvaCol is a visual problem-solving community: users draw solutions on a canvas, get AI review, publish to a public gallery, and others build on that work through clones and suggestions. This document sets the requirements for the first release (MVP), built as a single Spring Boot application with clear module boundaries so it can be split into services later.

Requirements are numbered (FR for functional, NFR for non-functional) so they can be referenced in tickets and tests. "Must" means required for the MVP; "Should" means planned but may follow shortly after.

Roles and key terms

A user's role is set per canvas: the same person can own one canvas and contribute to another.

Term	Meaning
Owner	The user who created the canvas. Only the owner can change the published canvas or accept suggestions.
Contributor	Any signed-in user who clones someone else's canvas to work on it.
Visitor	A user, signed in or not, who browses the gallery and views canvases.
Canvas	A drawing document: shapes, text and arrows, stored as scene JSON.
Version	A saved, immutable snapshot of a canvas. Every save creates a new version.
Published version	The version shown in the gallery. Only the owner sets it.

Term	Meaning
Clone	A contributor's own copy of a canvas, linked back to the original.
Suggestion	A request from a contributor asking the owner to review their clone and adopt it.
Recommendation	An AI-generated review or improved version of a canvas, returned by the LLM engine.

Functional requirements

All 14 original requirements are kept. Overlapping items (load, edit, save, get saved canvases) are merged, and applying an AI recommendation now creates a new version instead of overwriting the published copy, so nothing is ever lost.

Accounts

ID	Requirement	Priority
FR-01	A user can sign up with email and password and must verify their email before publishing.	Must
FR-02	A user can log in and log out. Login issues a JWT used for all API and WebSocket calls.	Must
FR-03	A user can reset a forgotten password by email link.	Must
FR-04	A user can set up multi-factor authentication (see NFR-01).	Must

Canvas editing

ID	Requirement	Priority
FR-05	A user can create a new canvas and becomes its Owner.	Must

ID	Requirement	Priority
FR-06	A user can load, edit and save their canvases. Every save creates a new version.	Must
FR-07	Unsaved changes are autosaved as a private draft every 30 seconds.	Should
FR-08	A user can list their saved canvases, separated into "Owned" and "Clones".	Must
FR-09	**Save** keeps a version private. **Submit** publishes the chosen version to the gallery.	Must
FR-10	An owner can delete their canvas. It leaves the gallery; existing clones keep their copy and attribution.	Should

Gallery

ID	Requirement	Priority
FR-11	Anyone can browse a gallery of published canvases. Each card shows the owner's published version.	Must
FR-12	Anyone can open a published canvas read-only, with its owner, credited contributors and clone count.	Must
FR-13	Every published canvas has a shareable link.	Must
FR-14	Users can search and filter the gallery by title and tag.	Should

Ownership and contributions

ID	Requirement	Priority
FR-15	Nobody but the owner can change a published canvas.	Must

ID	Requirement	Priority
FR-16	A contributor can clone a published canvas. The clone records its source canvas and version and shows "Cloned from @owner".	Must
FR-17	A contributor can edit their clone freely without affecting the original.	Must
FR-18	A contributor can send the owner a suggestion (their clone plus a short note) for review.	Must
FR-19	Owner and contributor can view a side-by-side diff that highlights added, removed and changed elements.	Must
FR-20	The owner can accept or reject a suggestion. Accepting creates a new version of the original and credits the contributor by name.	Must
FR-21	Each original canvas lists its clones, sorted by votes.	Should

Versions

ID	Requirement	Priority
FR-22	Users can view a canvas's version history with author, date and note for each version.	Must
FR-23	Users can open any older version read-only.	Must
FR-24	An owner can restore an older version. Restoring creates a new version; no version is ever deleted.	Should

AI review

ID	Requirement	Priority
FR-25	Any user editing a canvas (their own or a clone) can request an AI review, before or after saving.	Must

ID	Requirement	Priority
FR-26	Reviews run asynchronously. The user keeps editing and is notified when results are ready.	Must
FR-27	The LLM engine writes recommendations to the NoSQL store; Postgres stores a reference to each (recommendation ID, canvas, version, status).	Must
FR-28	The user sees recommendations as a critique (gaps, unnecessary complexity, alternatives) and, where available, a suggested revised canvas.	Must
FR-29	Applying a recommendation creates a new version of the canvas being edited. For an owner it reaches the gallery only when they submit it; for a contributor it stays in their clone.	Must
FR-30	AI reviews are limited per user per day to control cost.	Must

Voting and moderation

ID	Requirement	Priority
FR-31	Signed-in users can upvote or downvote a published canvas. A downvote requires a reason (unclear, incorrect, incomplete, low effort).	Should
FR-32	New canvases appear in a "New" tab for 7 days. After at least 20 votes, a canvas below 40% upvotes moves to "Needs work": hidden from the main gallery, not deleted, and the owner is notified with the reasons.	Should
FR-33	A revised version gets a fresh review window and can return to the gallery.	Should
FR-34	Users can report abusive or inappropriate content for review.	Must

Non-functional requirements

Targets are set for the MVP scale: about 1,000 registered users and 100 using the app at once. They should be revisited after launch.

ID	Area	Requirement	Priority
NFR-01	Security	Multi-factor authentication with an authenticator app (TOTP) is available to every user, with one-time recovery codes.	Must
NFR-02	Security	Passwords are hashed with bcrypt, all traffic uses HTTPS, and access tokens expire after 15 minutes (refresh tokens after 7 days).	Must
NFR-03	Security	Ownership is checked on the server for every write; the client is never trusted to enforce "owner only".	Must
NFR-04	Security	The LLM engine receives only canvas content. It never has access to credentials, emails or the user database.	Must
NFR-05	Collaboration	Invited users can co-edit the same canvas in real time. Edits appear for others within 1 second and simultaneous edits merge without loss (CRDT, e.g. Yjs).	Should
NFR-06	Performance	A canvas of up to 2,000 elements loads in under 2 seconds and saves in under 500 ms, for 95% of requests.	Must
NFR-07	Performance	A gallery page loads in under 1.5 seconds. AI review results arrive within 60 seconds for 95% of requests.	Must
NFR-08	Scalability	The app is stateless (sessions in JWT, data in databases) so more instances can run behind a load balancer without code changes.	Must
NFR-09	Maintainability	Built as a modular monolith with separate Auth, Canvas, Gallery and AI modules that talk only through interfaces, ready to split into services.	Must

ID	Area	Requirement	Priority
NFR-10	Availability	99.5% monthly uptime. Databases are backed up daily and a restore is tested monthly.	Must
NFR-11	Data integrity	Versions are immutable once saved, and a save either completes fully or not at all.	Must
NFR-12	Privacy	Drafts and unsubmitted versions stay private. Users can delete their account and personal data.	Must
NFR-13	Cost	LLM spend is tracked per user and per day, with an alert when it passes a set budget.	Must
NFR-14	Observability	Errors, request times and AI job failures are logged and monitored, with alerts on error spikes.	Should

Data storage

Postgres holds structured records and relationships; NoSQL (MongoDB) holds the drawings and AI output; object storage holds files.

Store	What it holds
Postgres	Users, credentials and MFA settings; canvas records (owner, title, tags, published version ID); version index; clone links; suggestions and their status; votes; references to AI recommendations.
NoSQL (MongoDB)	Canvas versions as scene JSON; AI recommendations (critique text and suggested canvas JSON).
Object storage	Thumbnails for gallery cards, uploaded images and exports.

Out of scope and open questions

Not in the MVP: timed challenges and competitions, zone-based live events, chat, mobile apps, and splitting into microservices.

- [ ] Should MFA be optional for all users or required for some (for example, accounts with many published canvases)?
- [ ] Can contributors see each other's clones before the owner accepts them, or only after?
- [ ] Does the owner approve who joins real-time co-editing (NFR-05), or can any signed-in user join a shared canvas?
- [ ] Which LLM provider and model will the AI review use, and what is the daily review limit per user (FR-30)?
- [ ] What are the exact moderation rules and who reviews reported content (FR-34)?